

Brennan Lee

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EDUCATION

University College London

Sep 2026 – Sep 2027

MSc Energy Systems and Data Analytics

London

- Will explore data analytics and modelling across energy systems, transport and the built environment

University of Bath

Oct 2022 – Jul 2026

BEng Chemical Engineering (with placement)

Bath

- Achieved 77.5% (1st Honours) in Year 1 and 2, with key principles such as Fluid Mechanics, Reaction Engineering and Process Safety
- Working towards IChemE chartership

WORK EXPERIENCE

Everoze

Sep 2024 – Jul 2025

Process & Systems Engineering Intern (Solar & Storage)

Bristol

- Reduced lender risk exposure across £500M+ in BESS and solar portfolios (650kWp to 360MWp) by producing 15+ technical due diligence reports covering site feasibility assessments, grid integration reviews and actionable risk mitigation recommendations
- Enabled solar site selection for two distilleries by performing RAG land assessments across 20+ land parcels to reduce development risk, applying site feasibility criteria such as grid proximity, environmental considerations and shading analysis
- Improved CAPEX and OPEX forecasting for BESS projects by building a dynamic cost benchmarking model in Excel using Solver-based optimisation, providing scalable process economics across varying project sizes to support data-driven client recommendations

PricewaterhouseCoopers

Aug 2023 – Sep 2023

Advisory Intern | Capital Projects & Infrastructure

Kuala Lumpur

- Strengthened a client's negotiation position on a £30 million airline merger deal by conducting technical and financial benchmarking across 4 case studies and evaluating reorganisation decisions and asset utilisation
- Compared airline merger case studies by collating and normalising financial and operational metrics (asset turnover, profit per available seat kilometres and aircraft utilisation), demonstrating analytical and multi-disciplinary skills applicable to engineering modelling

IOI Oleochemical

Jan 2021 – Mar 2021

Process engineering trainee

Penang

- Applied process safety and quality assurance principles in a manufacturing plant by observing and documenting HACCP monitoring across production lines, gaining exposure to HSE compliance, utility optimisation strategies and inter-department collaboration between quality control, assurance and procurement

ENGINEERING PROJECTS & TECHNICAL SKILLS

Sour water gas shift (Final Year Individual Project)

Apr – May 2026

- Delivered a fully specified plug flow sour water gas shift reactor, as part of the Fischer-Tropsch process, in a 70-page individual report using rigorous assumptions to model the reaction kinetics, temperature dependence and pressure drop across the reactor in MATLAB (1,000+ lines of code)
- Improved the process economics and sustainability of the initial design by optimising the operating temperature to catalyst weight according to multiple sensitivity analyses
- Demonstrated end-to-end process safety through a HAZID, P&ID with ratio, flow and temperature control loops and LOPA scenario, and developed a 3D AutoCAD engineering drawing based on a complete mechanical design of vessel geometry, wall thickness and auxiliary dimensions according to ASME and NACE-compliance standards

Sustainable Aviation Fuel (Final Year Design Project)

Feb – Apr 2026

- Advanced a concept-to-design sustainable aviation fuel (SAF) production route by coordinating a team of 8, translating an open-ended brief into a structured 135-page pre-FEED methodology (defining key assumptions

for complete mass and energy balance, equipment design, plant economics and sustainability considerations) within a 7-week timeline

- Optimised the production route by considering process economics, sustainability and ethical concerns by evaluating feedstock options, conversion route trade-offs, pinch analysis and lifecycle emissions against CORSIA compliance criteria, applying chemical engineering principles to assess technical, financial and ethical feasibility
- Strengthened project management and communication by introducing weekly planning sessions with defined milestones, coordinating multi-disciplinary inputs across process engineering design, equipment sizing and plant economics

Wastewater treatment in Morocco (Final Year Coursework)

Oct – Nov 2025

- Delivered a conceptual membrane bioreactor design, control strategy and process cost model within 4 weeks, demonstrating the ability to progress a concept through to a process design with a defined control architecture
- Created a model in MATLAB (550+ lines of code) that tracked the transmembrane pressure and concentrations of biomass, COD, and oxygen using mass balance ODEs and stoichiometric relationships, validating outputs against target product specifications

University Lab Programme

Oct 2024 – Jun 2024

12 technical reports across 36 hours of experimental lab work

- Demonstrated chemical engineering principles across plug flow reactor modelling, Bernoulli's principle and PID process control by processing raw experimental data using Excel and MATLAB, critically evaluating findings against scientific literature

ACTIVITIES

Biodevices without Borders

Oct 2025 – May 2026

Commercial Sensors team

- Contributed to the design and validation of an affordable commercial water quality sensor for rural communities by developing and executing experimental protocols to test pH, conductivity and dissolved oxygen sensors, ensuring reliable performance in various field deployment scenarios

Bath Malaysian Society

Oct 2023 – Jun 2024

Lead Treasurer

- Managed £3,500+ annual budget and secured 5 new sponsorships to deliver events to 200+ members, demonstrating financial accountability, negotiation and stakeholder management skills

IT SKILLS & PROFESSIONAL DEVELOPMENT

Microsoft Excel: Advanced data modelling, process cost benchmarking, CAPEX/ OPEX forecasting and Solver-based optimisation

MATLAB: Process and reaction simulation, ODE-based modelling and PID control for chemical engineering applications

Aspen Plus: Process simulation, unit operation optimisation for distillation, heat and reaction systems

Experienced in conducting process simulations and optimising various pre-FEED unit operations

AutoCAD: Creation of reactor mechanical design and plant layouts for processes

Certification: IChemE student member (working towards chartership), Capital and Financial Markets (Khan Academy) and Level 4 certificate (Institute of Banking and Finance)

INTERESTS

Gym (4 times a week), Cooking/baking (documenting my experimental side in my food blog), acoustic guitar (self-directed learning to challenge myself)